
GazeBot

Head-Gaze and Sip-and-Puff Shared Autonomy System for Assistive Robot Manipulation

Using Hello Robot Stretch 3 and Meta Quest 3

Project Proposal

McKechnie Family LIFE Home
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Duration: 8 Weeks (20 hrs/week, 160 total hours)

Location: LIFE Home, 75 Bailey Drive, Champaign, IL

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Executive Summary

This proposal outlines an 8-week research project to develop **GazeBot**, a hands-free, voice-free teleoperation system that enables individuals with severe motor disabilities to control the Hello Robot Stretch 3 mobile manipulator using only head movements and a sip-and-puff switch, mediated through a Meta Quest 3 XR headset.

The **target user population** includes individuals who are wheelchair-bound, have no functional use of their hands, and cannot speak—such as those living with advanced ALS, high-level spinal cord injuries (C1–C4), locked-in syndrome, or severe cerebral palsy. These individuals retain head movement and oral motor control, which GazeBot leverages as the sole control inputs.

The system employs a **shared autonomy paradigm**: the user provides high-level intent (selecting objects and destinations via head-gaze) while the Stretch 3 robot autonomously handles low-level motion planning, navigation, and grasp execution. This dramatically reduces the control bandwidth required from the user, making complex manipulation tasks achievable through minimal inputs.

This project directly aligns with the LIFE Home’s mission of enabling people of all ages and abilities to live fuller, healthier, and more autonomous lives, and builds upon the lab’s existing work with the Hello Robot Stretch and extended reality technologies.

Background and Motivation

The Problem

Approximately 5.4 million Americans live with paralysis, and over 3.6 million use wheelchairs (Christopher & Dana Reeve Foundation, 2024). Among these, individuals with conditions such as advanced ALS, C1–C4 spinal cord injuries, locked-in syndrome, and severe cerebral palsy face the most extreme limitations: they cannot use their hands, cannot speak, and are fully dependent on caregivers for basic activities of daily living (ADLs)—fetching objects, opening doors, getting food or water, and operating household items.

Current assistive robot interfaces overwhelmingly rely on hand-operated joysticks, touchscreens, or voice commands—all of which are inaccessible to this population. There is a critical gap in accessible robot control interfaces that work with **head movement and oral motor control only**.

Why Now?

Three converging technologies make this project timely:

1. **Hello Robot Stretch 3**—A compact, affordable (\$24,950) mobile manipulator specifically designed for home environments, with proven assistive applications (Olatunji et al., 2024). The LIFE Home already operates a Stretch robot for research.
2. **Meta Quest 3**—A \$500 XR headset with sub-degree, 90 Hz head tracking, color mixed-reality passthrough, and Bluetooth 5.2 connectivity. The LIFE Home’s Extended Reality Lab already houses Meta VR devices.
3. **Shared Autonomy Research**—Recent advances in learned latent actions (Jeon et al., 2020), gaze-based robot control (MIHRaGe, 2025; RaycastGrasp, 2025), and autonomous grasping (**stretch_ai**) enable high-level intent-based control that dramatically reduces the input bandwidth required from users.

Alignment with LIFE Home Research

This project directly extends several active LIFE Home initiatives:

- **STRETCH Project (NIH SBIR):** Refining the Stretch robot for older adults with cognitive and physical impairments. GazeBot extends this to users with severe motor disabilities.
- **ENHANCE Center (NIDILRR RERC):** Enhancing neurocognitive health and community engagement through technology.
- **CREATE VR Project:** Designing immersive XR interventions. GazeBot applies XR as an accessibility-first robot control interface.
- **Immersive Participatory Design Study (Olatunji, Mahajan et al., 2024):** Deployed Stretch in the home of Henry Evans (quadriplegic) for 4 weeks, finding 68% performance increase and 72% satisfaction improvement. GazeBot builds on this by creating an interface specifically designed for users who cannot use hands or voice.

Literature Review

Assistive Robot Teleoperation

Olatunji et al. (2024) placed a Stretch in the home of a quadriplegic older adult and his care partner for 4 weeks, finding a 68% increase in task performance through participatory design. The UW CREATE lab demonstrated that customized interfaces significantly improve performance for users with disabilities. The ALS Association and University of Pittsburgh have funded a \$400,000 project developing Stretch for people living with ALS.

Gaze-Based Robot Control

Recent work demonstrates the viability of gaze-based robot manipulation:

- **MIHRaGe (2025)** combines gaze tracking with mixed reality for hands-free object manipulation by mobility-impaired individuals.
- **RaycastGrasp (2025)** achieves >88% object detection confidence using stable gaze fixation, demonstrating natural gaze as a reliable assistive control modality.
- **Intent at a Glance (2026)** uses foundation models combined with gaze input for robotic intent inference.
- A **scoping review** (Frontiers, 2024) identified three gaze control approaches: telemanipulation with displays, directional gaze, and object-oriented gaze with 3D determination.

Shared Autonomy

Shared autonomy is essential for users with limited input bandwidth:

- **Jeon et al. (RSS 2020)** demonstrated that learned latent actions enable a 2-DoF joystick to control a 7-DoF robot arm, being both faster and preferred by users.
- **Admoni et al. (CMU, RSS 2015)** blended brain-computer interface control with autonomous assistance, specifically exploring sip-and-puff interfaces for motor-impaired users.
- **ASHA (2022)** applied human-in-the-loop reinforcement learning for assistive teleoperation.

XR Robot Teleoperation

- **OPEN TEACH (CoRL 2024)** provides open-source Meta Quest 3 teleoperation for Stretch at 90 Hz with 87% imitation learning success.
- **BEAVR (MIT, 2025)** achieves ≤ 35 ms latency VR teleoperation across multiple robot platforms.
- **XRoboToolkit (2025)** offers a cross-platform OpenXR framework for diverse robot control.

Gap in Literature

No existing system combines: (1) head-gaze control via XR headset, (2) sip-and-puff confirmation, (3) shared autonomy for grasp and navigation, and (4) the Stretch 3 mobile manipulator—specifically designed for users who cannot use hands or voice. **GazeBot fills this gap.**

Research Objectives

Primary Objective

Develop and evaluate a hands-free, voice-free teleoperation system that enables individuals with severe motor disabilities to perform activities of daily living using the Hello Robot Stretch 3, controlled entirely through head movements and sip-and-puff input via a Meta Quest 3 XR headset.

Secondary Objectives

1. **Design a head-gaze control interface** that maps Quest 3 head tracking to robot camera control, navigation goals, and object selection within a mixed-reality environment.
2. **Integrate sip-and-puff input** as the sole confirmation/selection mechanism, connected via Xbox Adaptive Controller over Bluetooth to the Quest 3.
3. **Implement a shared autonomy layer** where the user provides high-level intent and the robot autonomously executes navigation, grasp planning, and manipulation.
4. **Develop task presets** for common ADLs: fetching objects, opening doors, clearing paths, and retrieving food/water.
5. **Evaluate system performance** through structured task trials measuring success rate, completion time, and subjective usability in the LIFE Home environment.

System Architecture

Overview

The system consists of three main components connected over Wi-Fi:

User Layer (Wheelchair) Meta Quest 3 (head-gaze, 90 Hz) + Sip-and-Puff via Xbox Adaptive Controller (BT)
↓
Application Layer (Quest 3) Custom Unity/OpenXR App: gaze ray casting, mode management, MR overlay, audio feedback
↓ Wi-Fi (WebRTC + ROS2 Bridge)
Robot Layer (Stretch 3) ROS2 + Nav2 + stretch_ai + shared autonomy decision layer

Input Mapping

Input	Action
Head rotation (yaw/pitch)	Gaze ray direction—selects targets in robot camera view
Soft sip	Confirm / Select
Soft puff	Cancel / Back
Hard sip	Switch control mode
Hard puff	Emergency stop—immediately halts all robot motion

Control Modes

Mode	Head-Gaze Maps To	Sip Action
Look	Robot head pan/tilt (camera)	Set viewpoint
Navigate	Target destination in environment	Go to target (Nav2)
Manipulate	Object selection via gaze ray	Grasp/place (autonomous)
Menu	UI option navigation	Select preset task

Shared Autonomy Pipeline

1. **User looks at object** → head-gaze ray highlights detected object in MR overlay
2. **User sips to confirm** → `stretch_ai` identifies object via YOLO/SAM
3. **Robot plans grasp** → trajectory preview shown in MR view
4. **User sips to approve** → robot executes autonomous grasp
5. **User looks at destination** → head-gaze indicates placement location
6. **User sips to confirm** → robot navigates (Nav2) and places object

Technology Stack

Component	Technology
XR Application	Unity 2022 LTS + Meta XR SDK + OpenXR
Head Tracking	Meta Quest 3 inside-out tracking (6DoF, 90 Hz)
Sip-and-Puff	Origin Instruments or generic switch + Xbox Adaptive Controller
Robot Control	ROS2 Humble + <code>stretch.body</code> Python SDK
Navigation	Nav2 (ROS2) with RPLIDAR A1
Object Detection	<code>stretch.ai</code> (YOLO + SAM)
Grasp Planning	<code>stretch.ai</code> / FUNMAP whole-body grasping
Video Streaming	WebRTC (via <code>stretch_web_teleop</code> infrastructure)
Communication	ROSBridge / WebSocket over Wi-Fi 6E
Audio Feedback	Bluetooth earbuds with confirmation tones

Methodology

Development Approach

The project follows an iterative, incremental development approach with weekly deliverables. Each week produces a functional system increment that can be demonstrated and tested in the LIFE Home environment.

Evaluation Plan

Quantitative Metrics

- Task success rate (%) across 5 standard ADL tasks
- Task completion time (seconds)
- Number of user inputs (sips/puffs) required per task
- Robot autonomy utilization rate (% of subtasks handled autonomously)
- System latency (head movement to robot response, ms)

Qualitative Metrics

- NASA-TLX workload assessment (adapted for head-gaze input)
- System Usability Scale (SUS)
- Semi-structured interview on perceived usefulness and frustrations

Evaluation Tasks (Conducted in LIFE Home)

1. Fetch a specified object from a table and bring to user
2. Navigate robot to a different room
3. Open a cabinet door and retrieve an item
4. Pick up an object from the floor
5. Execute a preset “get water” task sequence

Protocol: 5 tasks $\times$ 3 trials each = 15 trials per evaluation session. Conducted in the LIFE Home’s home simulation area with observation cameras recording for analysis.

Timeline (8 Weeks)

Week 1: Environment Setup and Baseline (20 hrs)

Task	Hours
Configure Stretch 3 with ROS2, <code>stretch_body</code> , <code>stretch_ai</code> in LIFE Home	6
Set up Unity development environment with Meta XR SDK for Quest 3	6
Get OPEN TEACH baseline running (Quest 3 → Stretch, hand-tracking mode)	5
Literature deep-dive: MIHRaGe, Learned Latent Actions, RaycastGrasp papers	3

Milestone: Stretch controllable via Quest 3 hand tracking (OPEN TEACH baseline).

Week 2: Head-Gaze Control System (20 hrs)

Task	Hours
Build Unity app reading Quest 3 head pose (6DoF) via OpenXR at 90 Hz	4
Implement head-gaze ray casting into robot camera view	4
Stream Stretch head camera into Quest 3 via WebRTC	6
Map head rotation to robot head pan/tilt—“look through robot’s eyes”	4
Tune sensitivity, dead zones, smoothing filters for comfort	2

Milestone: User sees through robot’s eyes and controls camera with head movements.

Week 3: Sip-and-Puff Integration and Mode System (20 hrs)

Task	Hours
Connect SNP switch to Xbox Adaptive Controller, pair via Bluetooth	3
Map sip/puff signals to Unity app actions (soft/hard differentiation)	5
Implement 4 control modes: Look, Navigate, Manipulate, Menu	8
Build MR heads-up display showing mode, battery, connection status	4

Milestone: Full hands-free input—head-gaze + sip/puff controls the robot in all modes.

Week 4: Autonomous Navigation (20 hrs)

Task	Hours
Integrate Nav2—user looks at target, sips to set navigation goal	6
Convert head-gaze ray + depth image into 3D world coordinate	6
Show navigation path preview in MR view before user confirms	4
Test navigation throughout LIFE Home rooms	4

Milestone: User looks at a point in the LIFE Home, sips, and robot navigates there.

Week 5: Shared Autonomy Manipulation (20 hrs)

Task	Hours
Integrate object detection pipeline (<code>stretch_ai</code> YOLO/SAM)	4
Head-gaze ray highlights detected objects with visual feedback	4
Sip to confirm → trigger autonomous grasp planning	6
Show grasp preview in MR → sip to execute	3
Implement “bring to me”—robot grasps then navigates back to user	3

Milestone: User looks at an object, sips twice, robot picks it up and brings it over.

Week 6: Task Presets and Audio Feedback (20 hrs)

Task	Hours
Build preset task library: “Fetch object”, “Open door”, “Clear path”, “Get water”	8
Implement head-gaze menu for selecting preset tasks	3
Add audio feedback: confirmation tones, error alerts, task status	4
Refine emergency stop—hard puff instantly halts all motion	2
End-to-end integration testing	3

Milestone: Complete system with preset ADL tasks, audio feedback, and safety stops.

Week 7: Evaluation and Testing (20 hrs)

Task	Hours
Structured evaluation: 5 tasks $\times$ 3 trials, record success/time/inputs	8
Baseline comparison with <code>stretch_web_teleop</code> (keyboard/mouse)	4
Administer NASA-TLX and SUS questionnaires	2
Data analysis: success rates, completion times, workload scores	4
Fix critical usability issues discovered during evaluation	2

Milestone: Quantitative evaluation data demonstrating system viability.

Week 8: Documentation, Demo, and Dissemination (20 hrs)

Task	Hours
Record professional demo video (2–3 min) in LIFE Home	4
Write technical report with architecture, results, and future work	6
Clean and document codebase, push to GitHub with README	4
Prepare presentation for LIFE Home team / lab meeting	3
Draft conference paper outline (ASSETS 2026 or HRI 2027)	3

Milestone: Demo video, technical report, public GitHub repository, paper outline.

Expected Outcomes and Deliverables

Primary Deliverables

1. **GazeBot Software System**—Open-source Unity application + ROS2 nodes enabling head-gaze + sip-and-puff control of Stretch 3 via Quest 3
2. **Technical Report**—Detailed documentation of architecture, implementation, and results
3. **Demo Video**—2–3 minute professional demonstration recorded in the LIFE Home
4. **Evaluation Data**—Quantitative task performance metrics and subjective usability scores
5. **GitHub Repository**—Fully documented, reproducible codebase

Expected Performance Targets

Metric	Target
Task success rate (fetch object)	> 80%
Task completion time (fetch object)	< 120 seconds
System latency (head-to-robot)	< 200 ms
Emergency stop response time	< 500 ms
System Usability Scale score	> 68 (above average)

Publication Potential

- **ACM ASSETS 2026**—Conference on Computers and Accessibility
- **HRI 2027**—International Conference on Human-Robot Interaction
- **CHI 2027**—ACM Conference on Human Factors (Late-Breaking Work)
- **Frontiers in Robotics and AI**—Special topic on assistive robotics

Required Resources

Available at LIFE Home

Resource	Status
Hello Robot Stretch (mobile manipulator)	Available (bookable)
Meta Quest / Oculus VR headset	Available in Extended Reality Lab
Home simulation area	Available for testing
Observation cameras (6× Axis M5525-E, 1080p)	Available for evaluation recording
Wi-Fi infrastructure	Available

To Be Procured (Estimated Budget: \$200–\$400)

Item	Est. Cost
Xbox Adaptive Controller	\$99
Sip-and-puff switch (3.5mm jack)	\$50–150
Gooseneck mount for SNP tube	\$20–40
Bluetooth earbuds (audio feedback)	\$30–50
Total	\$200–340

Software (All Free / Open-Source)

Unity 2022 LTS (Free Education), Meta XR SDK (Free), ROS2 Humble (Apache 2.0), `stretch_ai` / `stretch_body` / `stretch_ros2` (Apache 2.0 / MIT), OPEN TEACH (Open source).

Risk Assessment and Mitigation

Risk	Likelihood	Impact	Mitigation
OPEN TEACH setup complexity	Medium	High	Fallback: build on <code>stretch_web_teleop</code>
SNP + Xbox controller latency	Low	Medium	Alternative: Augmental MouthPad or cheek switch
Autonomous grasping unreliable	Medium	High	Reduce to navigation-focused tasks; add manual override
Head-gaze too coarse for selection	Medium	Medium	Add dwell-time confirmation + zoom mode
Wi-Fi latency in LIFE Home	Low	Medium	Use Wi-Fi 6E direct; fallback to Ethernet
User fatigue from head-gaze	Medium	Medium	Rest prompts; design for short task bursts
Stretch availability conflicts	Low	Medium	Book dedicated slots; use MuJoCo simulation

Broader Impact and Future Directions

Immediate Impact

GazeBot would demonstrate that individuals with the most severe motor disabilities can independently perform household tasks through assistive robot teleoperation—a capability that currently does not exist in any commercially or academically available system.

Future Research Directions

1. **Longitudinal in-home deployment**—Following the participatory design methodology of Olatunji et al. (2024), deploy GazeBot in the home of an individual with severe motor disabilities for multi-week evaluation.
2. **Eye tracking integration**—When Meta Quest 4 (expected 2027–2028) introduces built-in eye tracking, GazeBot can leverage eye-gaze for faster, more precise object selection.
3. **Adaptive autonomy**—Machine learning system that learns user preferences and adjusts autonomy level over time.
4. **Wheelchair integration**—Unified interface where head-gaze + SNP inputs control both the power wheelchair and the Stretch robot via Tecla-e hub.
5. **AAC integration**—Combine robot control with augmentative and alternative communication through the same interface.
6. **Multi-robot coordination**—GazeBot controlling multiple Stretch robots for complex household scenarios.

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